23CSE202-DBMS

SQL BASICS

RECAP

- Introduction SQL , MYSQL
- SQL-Data types, operators,
- SQL commands-Create, Insert, Select

To know the table details

TABLE DESCRIPTION

- Describes the schema of the table
- **DESC table_name;**
- Eg: DESC employee;

DELETE

To delete the particular row/tuple

- **DELETE FROM table_name WHERE condition;**

Eg:

delete from employee where empid=200;

delete from employee where empid=200 and salary=2300;

TRUNCATE TABLE

TRUNCATE TABLE

- **Used to delete the data inside a table, but not the table itself.**

TRUNCATE TABLE *table_name*;

- Eg:
- TRUNCATE TABLE *employee*;

DROP TABLE

DROP TABLE

Drops the existing table

DROP TABLE *table_name*;

Eg:

DROP TABLE *employee*;

About **Insert** Commands

To insert a record in all the columns

insert into table_name values (value1, value2, value3 …….);

e.g.

insert into emp values(1,’teena’,45678,’manager’);

To insert multiple records in all the columns

insert into table_name values (value1, value2, value3 …….), (value1, value2, value3 …….);

e.g.

insert into emp values(3,’leena’,43000,’clerk’),(4,’meena’,47000,’analyst’);

About Insert Commands

To insert records in to specific columns

insert into table_name(column_name1,column_name2) values (value1, value2);

e.g.

insert into emp(name, empid) values(‘sheena’,2);

To Insert Null Values

e.g.

insert into emp values(3,’heena’,50000,NULL);

Select-relational operator- > , < ,<= , >= ,=(equal to), <>(not equal to)

Display the employee information whose name is not teena from the table emp

- e.g.
- **select * from emp where name<>“teena”;**

To display name and designation of those employees ,from the table emp, whose salary is more than 48000.

- **select name,designation from emp where salary>48000;**

Select- where command- logical operator- and, or , not

Display the employee information whose salary is more than 48000 and name is not teena from the table emp

- **select * from emp where name<>“teena” and salary >48000;**

To display name and designation of those employees ,from the table emp, whose salary is more than 48000 or designation is clerk.

- **select name,designation from emp where salary>48000 or designation =‘clerk’;**

Select, where command-
between keyword- condition based on a range

syntax:

- **select */ column_name from table where column_name between value1 and value2;**

Display the employee information whose salary is in the range of 45000 to 55000 from the table emp

- select * from emp where salary between 45000 AND 55000;

Display the employee information whose salary is not between 45000 to 55000 from the table emp

- select * from emp where salary not between 45000 AND 55000;

Select, where command-
in keyword- condition based on a list

syntax:

- **select */ column_name from table where column_name in (value1, value2 …);**

Display only manager and clerk employee information from the table emp

select * from emp where designation **in**(‘manager’, ‘clerk’);

OR

select * from emp where designation =‘manager’ or designation =‘clerk’;

Select, where command-
in keyword- condition based on a list

Display all designation except manager and clerk from the table emp

select * from emp where designation **not in**(‘manager’, ‘clerk’);

OR

select * from emp where designation<>’manager’ or designation <>‘clerk’;

Select, where command-
Like operator- condition based on pattern matches

percent(%)- to match any substring

syntax:

select */ column_name from table where column_name like ‘ % ‘;

e.g.

To list those employee information whose name is **starting from** ‘m’ from the table emp

- select * from emp where name like **‘m%’**;

Display employee information whose name is **ending with** ‘a’ from the table emp

- select * from emp where name like **‘%a’**;

Select, where command-
Like operator- condition based on pattern matches

underscore(_)- to match any single character

syntax:

select */ column_name from table where column_name like ‘ _ ‘;

e.g.

- To list those employee information whose name’s length is 5 from the table emp

  select * from emp where name like ‘_ _ _ _ _’;

- Display employee information whose name is 5 character long and second character must be ‘e’ from the table emp

  select * from emp where name like ‘_e_ _ _’;

- Display employee information whose name’s second character must be ‘e’ from the table emp

  select * from emp where name like ‘_e%’;

Example

- SELECT * FROM employee WHERE name LIKE 'A%’;
- SELECT * FROM employee WHERE name LIKE '%n’;
- SELECT * FROM employee WHERE name LIKE '%ar%’;
- SELECT * FROM employee WHERE name LIKE '_o%’;
- SELECT * FROM employee WHERE department LIKE 'Eng%’;
- SELECT * FROM employee WHERE name LIKE '_e_____' AND CHAR_LENGTH(name) = 10;

Select where command-
NULL operator- searching for NULL

syntax:

select */ column_name from table where column_name is NULL;

- e.g.
- To list those employee information whose designation is not filled.

  select * from emp where designation is **NULL**;
- Display employee information where designation is not empty.

  select * from emp where designation is **not NULL**;

ALL keywordshow all values (retains duplicates values)

syntax:

select all column_name from table;

or

select * from table;

e.g.

(i). To list the designation from the table emp

select **all** designation from emp;

Distinct keyword- show only unique values

syntax:

select distinct column_name from table;

e.g.

To list the different types of designation from the table emp

select **distinct** designation from emp;

Current date and time

- select **curdate();** - to show system current date

- select **curtime();** - to show system current time

Insert data into another table

Syntax

Insert into newtable_name select * from table_name where condition ;

e.g.

- Add those employee information into table emp2 where salary is less than 50000;
- **insert into emp2 select * from emp where salary<50000;**

Summary

Operators in The WHERE Clause

Operator	Description
=	Equal
<>	Not equal. Note: In some versions of SQL this operator may be written as !=
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal
BETWEEN	Between an inclusive range
LIKE	Search for a pattern
IN	To specify multiple possible values for a column

UPDATE

UPDATE

- The UPDATE statement is used to modify the existing records in a table.

UPDATE *table_name* SET *column1 = value1, column2 = value2, ...* WHERE *condition*;

Eg:

- UPDATE employee SET Email ='c_gokul@gmail.com' WHERE empid=200;

Note:
It Update not working execute the following command
set sql_safe_updates=0;

UPDATE…

- Increase the salary of all employee by 200;

update emp where salary= salary +200;

- Decrease the salary of all employee by 2%;

update emp where salary= salary –salary *0.02;

- Increase the salary of those employee by 200 whose salary is less than 40000;

update emp where salary= salary +200 where salary <40000;

ALTER TABLE

- The SQL ALTER TABLE statement is a powerful tool that allows you to modify the structure of an existing table in a database.

  - The ALTER TABLE statement in SQL is used to modify an existing table structure in a database without losing any data.
  - It allows you to add, remove, or modify columns, change data types or apply constraints to improve data integrity and ensure that the table meets evolving business requirements.
  - It allows for structural changes like adding new columns, modifying existing ones, deleting columns and renaming columns within a table.

ALTER TABLE

- Syntax:

ALTER TABLE table_name

[ADD | DROP | MODIFY] column_name datatype;

Key Terms

table_name refers to the name of the table you want to modify.

ADD is used to add a new column.

DROP is used to remove an existing column.

MODIFY is used to change the datatype or definition of an existing column.

Alter –ADD Column

- ALTER TABLE - ADD Column

- To add a column in a table, use the following syntax:

- **ALTER TABLE *table_name* ADD *column_name datatype;***

Eg: ALTER TABLE employee ADD Email varchar(255);

UPDATE employee SET email = 'alice.johnson@example.com’ WHERE empid = 1;

Alter- DROP column

- To delete a column in a table, use the following syntax (notice that some database systems don't allow deleting a column):

- **ALTER TABLE *table_name* DROP COLUMN *column_name*;**

- Eg: ALTER TABLE employee DROP COLUMN Email;

Alter-Modify

- To modify the size of a particular column into existing table;

- Alter table emp modify email varchar(100);

ALTER– Rename -to change the name of the column

Syntax:

ALTER TABLE table_name RENAME COLUMN old_column_name TO new_column_name;

Eg:

alter table emp rename empname to EName;

ORDER BY

- The ORDER BY keyword is used to sort the result-set in ascending or descending order.
- syntax

select * /columm_names from table name where condition

order by column_name asc/desc

- asc for ascending.
- desc for descending.
- Default is ascending order.

ORDER BY

- Eg:
- To display employee's name in descending order

  SELECT * FROM employee ORDER BY name ASC;

  SELECT * FROM employee ORDER BY salary DESC;

- To display employee information in descending order of their name where salary is more than 5000

  SELECT * FROM employee WHERE salary > 50000 ORDER BY name DESC;

Numeric Functions

Function Name	Examples	Return Value
ABS (x)	ABS (1)	1
	ABS (-1)	-1
CEIL (x)	CEIL (2.83)	3
	CEIL (2.49)	3
	CEIL (-1.6)	-1
FLOOR (x)	FLOOR (2.83)	2
	FLOOR (2.49)	2
	FLOOR (-1.6)	-2
ROUND (x, y)	ROUND (125.456, 1)	125.4
	ROUND (125.456, 0)	125
	ROUND (124.456, -1)	120
TRUNC (x, y)	TRUNC (140.234, 2)	140.23
	TRUNC (-54, 1)	54
	TRUNC (5.7)	5
	TRUNC (142, -1)	140

Aggregate functions

- An aggregate function is a function that performs a calculation on a set of values, and returns a single value.
- The most commonly used SQL aggregate functions are:
  - avg(),
  - sum(),
  - max(),
  - min(),
  - count(column_name)-Count returns the number of rows present in the table either based on some condition or without condition.
  - count(distinct)

Aggregate functions

- Eg
- Average salary for all employee

  SELECT AVG(salary) FROM employee;

- Average salary for all employee whose title is equal to 'Programmer‘

  SELECT AVG(salary) FROM employee WHERE title = 'Programmer';

Aggregate functions

- To display the number of Employees
    - Select Count(*) FROM employee;
    - select count(Age) from pp;
    - select count(distinct Age) from pp;

Group By

- The GROUP BY statement groups rows that have the same values into summary rows, like "find the number of customers in each country".

- The GROUP BY statement is often used with aggregate functions (COUNT(), MAX(), MIN(), SUM(), AVG()) to group the result-set by one or more columns.

- Syntax
- **select * /columm_name(s) from table name where condition group by column_name**
- * Group By clause will always come at the end.

Group By

- To show name and sum of the salary of employees according to their department.

  SELECT department, SUM(salary) AS total_salary FROM employee
  GROUP BY department;

  - Group employees by department and count how many employees are in each.
- **SELECT department, COUNT(*) AS total_employees FROM employee GROUP BY department;**

Having clause

- It is used to give more precise condition for a statement.

- It is used to mention condition in Group based SQL functions, just like WHERE clause.
- The HAVING clause was added to SQL because the WHERE keyword cannot be used with aggregate functions.

Having clause

- Syntax

select * /columm_name(s) from table name where condition

group by column_name having condition

- Eg
- To show name and sum of the salary of employees of whose designation count is more than 2.

SELECT name,sum(salary) from employee group by department where count(*) >2;

SQL Alias

- Alias is used to give an another name to a table or a column.

Alias name to table

- Syntax
- **SELECT column-name from table-name table_alias-name;**
- Example

  SELECT * from Employee_detail ed;

Syntax: Alias name to column

- **SELECT column-name “alias-name “ from table-name;**

  SELECT customer_id “cid” from Emp;

DIFFERENCES

DDL	DML
Data definition language	Data manipulation language
Create	Insert
Alter	Update
Drop	delete

ALTER	UPDATE
DDL command	DML command
It is used to add /delete/change name of a particular column	It is used to change/modify the value(s) in particular column(s)
alter table tablename add/modify/change column_name datatype(size)	update tablename set column_name=value where column_name=value

DROP	DELETE
DDL command	DML command
It is used to delete the table or database permanently	It is used to delete a particular record(s) from the table
drop table table_name	delete from table_name where condition

WHERE	HAVING
Where- Where clause is used to specify condition on single row.	having- It is used to mention condition in Group
Where clause is used mostly with Select, Update and Delete command/query	Having clause is used only with group by clause

ORDER BY	GROUP BY
It is used to arrange records in a particular order(asc/desc)	It is used to group together similar types of information
select */column_name from table_name order by column_name asc/desc	**Select */column_name from table_name group by column_name**

1. Display the details of ‘NITA’ and ‘ANU’ from the instructor table.
2. Display the details of all instructors whose age ranges between 23 and 100.
3. Display the details of all instructors whose age is not 25
4. Display the details of instructor as the **ascending order** of salary
5. Display the instructor details with the following conditions:
    1. Department name should be in descending order
    2. Within each department salary should be in ascending order

ID	NAME	AGE	DEPT_NAME	SALARY
E01	NITA	23	CSE	40000
E02	VISHAL	29	MECHANICAL	45000
E03	ANU	30	EEE	35000
E04	ABHISHEK	35	CSE	50000
E05	ARUN	25	EEE	50000
E06	AJITH	32	CSE	38000
E07	PREMA	45	PHYSICS	60000
E08	RAJANI	48	PHYSICS	55000